+Naive Bayes Classifier

Naive bayes is a ML classification algorithm that predicts the category of a data point using probability.
It assume that all features are independent of each other.
Ex: best use in spam filtering, document categorisation, sentiment analysis

Naive bayes use **Bayes's Theorem** to classify data based on the probabilities of different classes given the features of the data. It is used mostly in high-dimensional text classification.

Naive bayes's Characteristic:
-Very few number of parameters --> faster than other classification algorithm
-It assumes that each feature contributes to the predictions with no relation between each other

Why is it called Naive Bayes?
Because it assumes the presence of one feature does not affect other features.

Assumption of the Naive Bayes

-**Feature independence**: when we are trying to classify sth, each feature or piece of information in the data does not affect any other feature.
-**Continuous features are normally distributed**: If a feature is continuous, then it is assumed to be normally distributed within each class.
-**Discrete features have multinomial distributions**: If a feature is discrete, then it is assumed to have a multinomial distribution within each class.
-**Features are equally important:** all feature are assumed to contribute equally to the prediction of the class label.
-**No missing data**: the data should not contain any missing values.

Bayes's Theorem

$$P(y|X) = \frac{P(X|y) \cdot P(y)}{P(X)}$$

Where:

- $P(y|X)$: Posterior probability, probability of class y given features X
- $P(X|y)$: Likelihood, probability of features X given class y
- $P(y)$: Prior probability of class y
- $P(X)$: Marginal likelihood or evidence

The "naive" in Naive Bayes comes from the assumption that all features are independent given the class. That is:

$$P(x_1, x_2, ..., x_n|y) = P(x_1|y) \cdot P(x_2|y) \cdots P(x_n|y)$$

Thus, Bayes' theorem becomes:

$$P(y|x_1, ..., x_n) = \frac{P(y) \cdot \prod_{i=1}^{n} P(x_i|y)}{P(X)}$$

Since the denominator is constant for a given input, we can write:

$$P(y|x_1, ..., x_n) \propto P(y) \cdot \prod_{i=1}^{n} P(x_i|y)$$

We compute the posterior for each class y and choose the class with the highest probability:

$$\hat{y} = \arg\max_y P(y) \cdot \prod_{i=1}^{n} P(x_i|y)$$

This becomes our Naive Bayes classifier.

Naive Bayes for Continuous Features
Continuous Feature = is a **variable** that can take any value within a **range** (including decimal/fraction values)
It comes from a **continuous domain**, usually measured rather than counted.

For continuous features, we assume a Guassian Distribution

$$P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

Type of Naive Bayes Model

1. **Gaussian Naive Bayes**: Continuous values associated with each feature are assumed to be distributed according to a gaussian distribution.

2. **Multinomial naive Bayes**: when features represent the frequency of terms (word count) in a doc. Commonly applied in text classification
3. **Bernoulli Naive Bayes**: binary feature, where each feature indicates whether a word appears or not in a document. It is suited for scenarios where the presence or absence of terms is more relevant than their frequency. Both models are widely used in document classification tasks